This procedure is applicable to below models:

V810 STD	√ S1	√ S2
V810 XXL	√ S1	√ S2
V810 S2EX	√	
V810 Mini	☐	

REVISION HISTORY

Rev	Effective Date	Description	Doc Originator
00	7 July 2017	First Release	Muhammed Shameem

Content

A. Inner Barrier Cylinder Replacement

I. Inner Barrier Cylinder Removal

1. Close Air Pressure Regulator
2. Remove 2 x Tubing as shown in Figure 1

   Remarks: Perform label marking at respective tubing upon remove

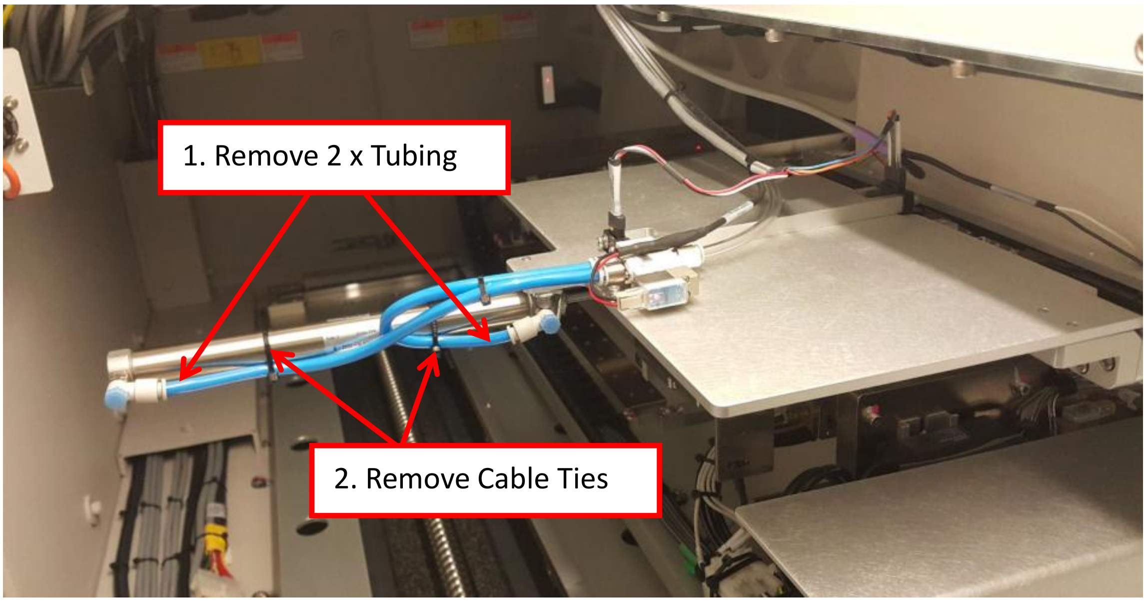

Figure 1 Inner Barrier Cylinder Tubing

3. Remove 2 x Speed Controller by using adjustable spanner as shown in Figure 2 & 3

   Remarks: Speed Controllers require to re-use upon replacement

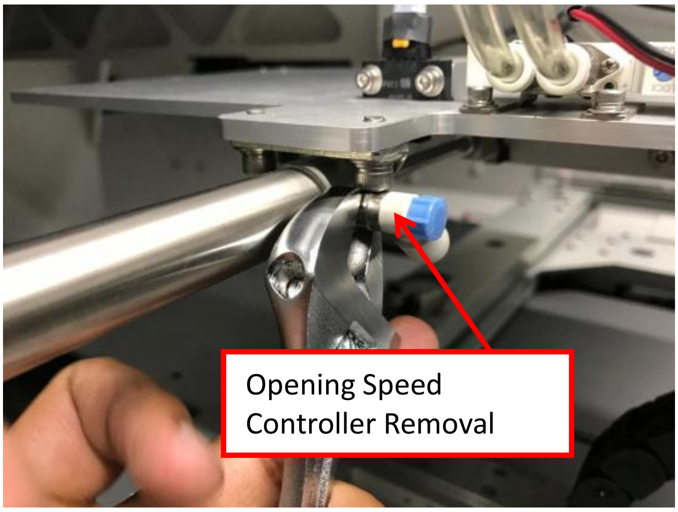

Figure 2 Opening Speed Controller Removal

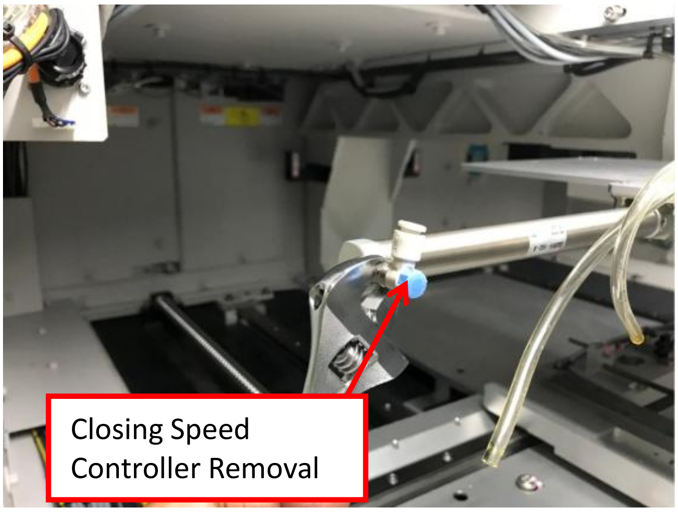

Figure 3 Closing Speed Controller Removal

4. Loosen 2 x Screws as shown in Figure 4.

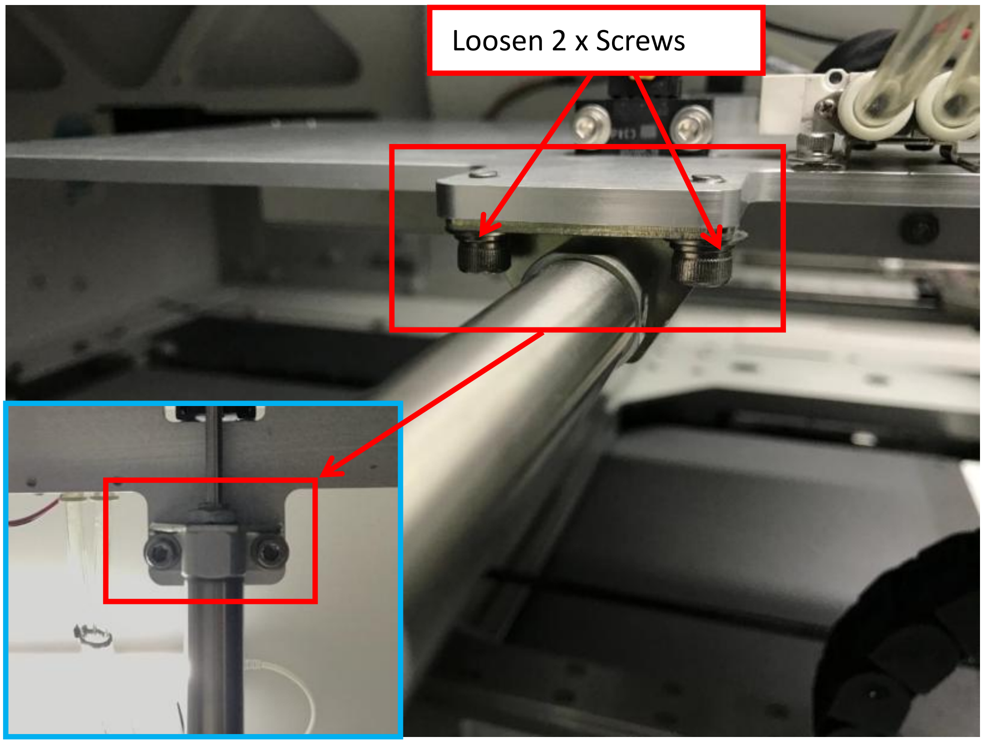

Figure 4 Inner Barrier Cylinder Removal

5. Gently release Double Knuckle Joint from Drive Bar`s hook as below Figure 5.

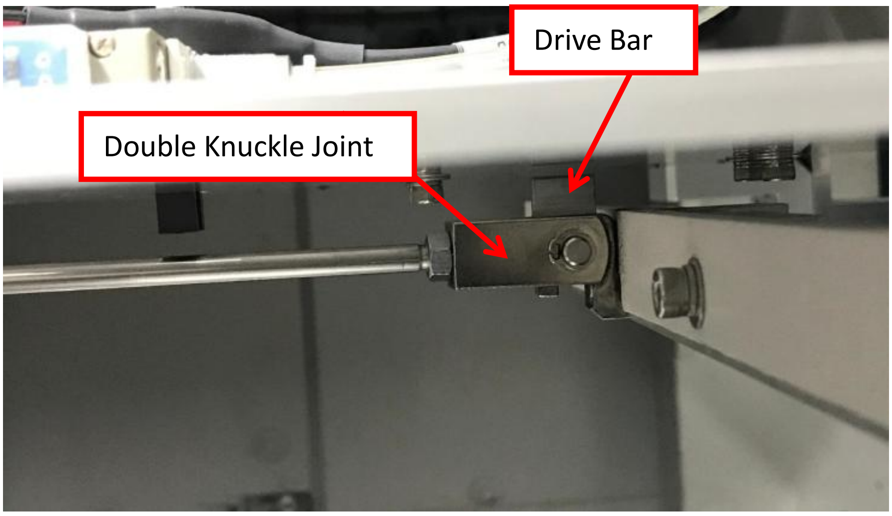

Figure 5 Inner Barrier Cylinder Removal

6. Remove Inner Barrier Module from machine as shown in Figure 6

Figure 6 Inner Barrier Cylinder Module

7. Remove below item from Inner Barrier Cylinder Module:
    i. Foot Bracket & Nut
    ii. Double Knuckle Joint & Nut
    iii. SMC Cylinder

Remarks:Foot Bracket ,Double Knuckle Joint & Nut will require to re-use upon replacement

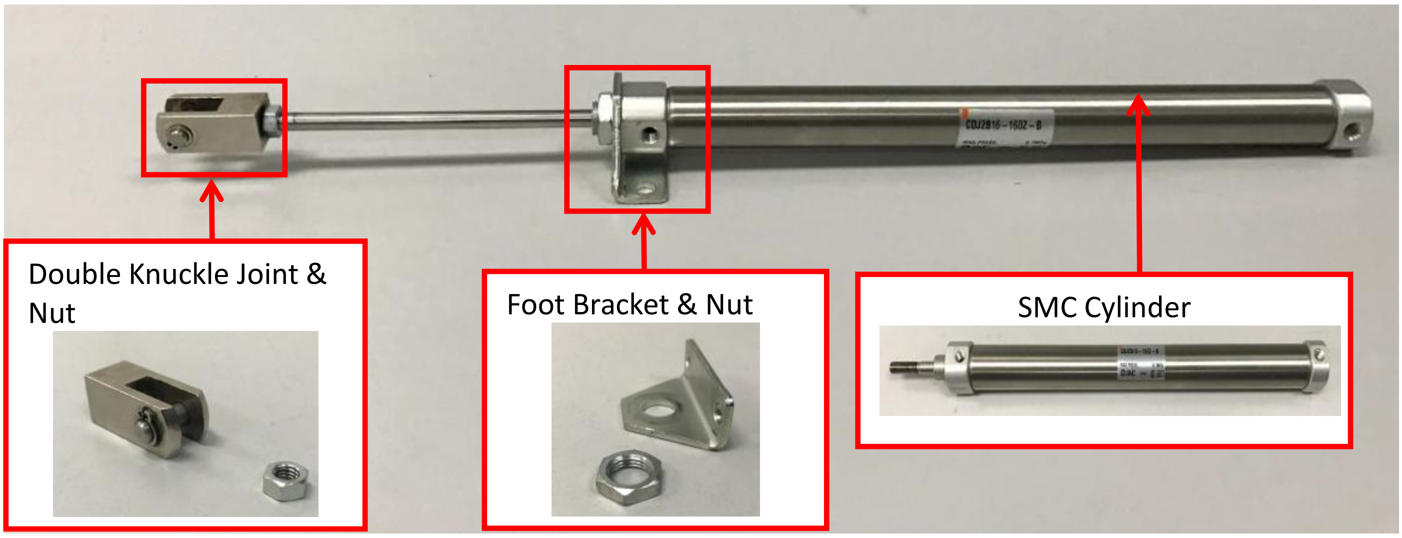

Figure 7 Inner Barrier Cylinder Module

II. Inner Barrier Cylinder Replacement

1. Prepare below listed item for Inner Barrier Cylinder replacement

   i. Double Knuckle Joint & Nut (Figure 8)

Figure 8 Double Knuckle Joint & Nut

   ii. Foot Bracket & Nut (Figure 9)

Figure 9 Foot Bracket & Nut

   iii. SMC Cylinder (Figure 10)

   - PN : 2810-0003 (For NON VM Feature)
   - PN : 2810-0034 (For VM Feature)

Figure 10 SMC Cylinder

2. Assemble all lower level part as shown in Figure (8,9 and 10) into Inner Barrier Cylinder Module as shown in Figure 10

   Remarks:Require to refer to below parts orientation upon assembly

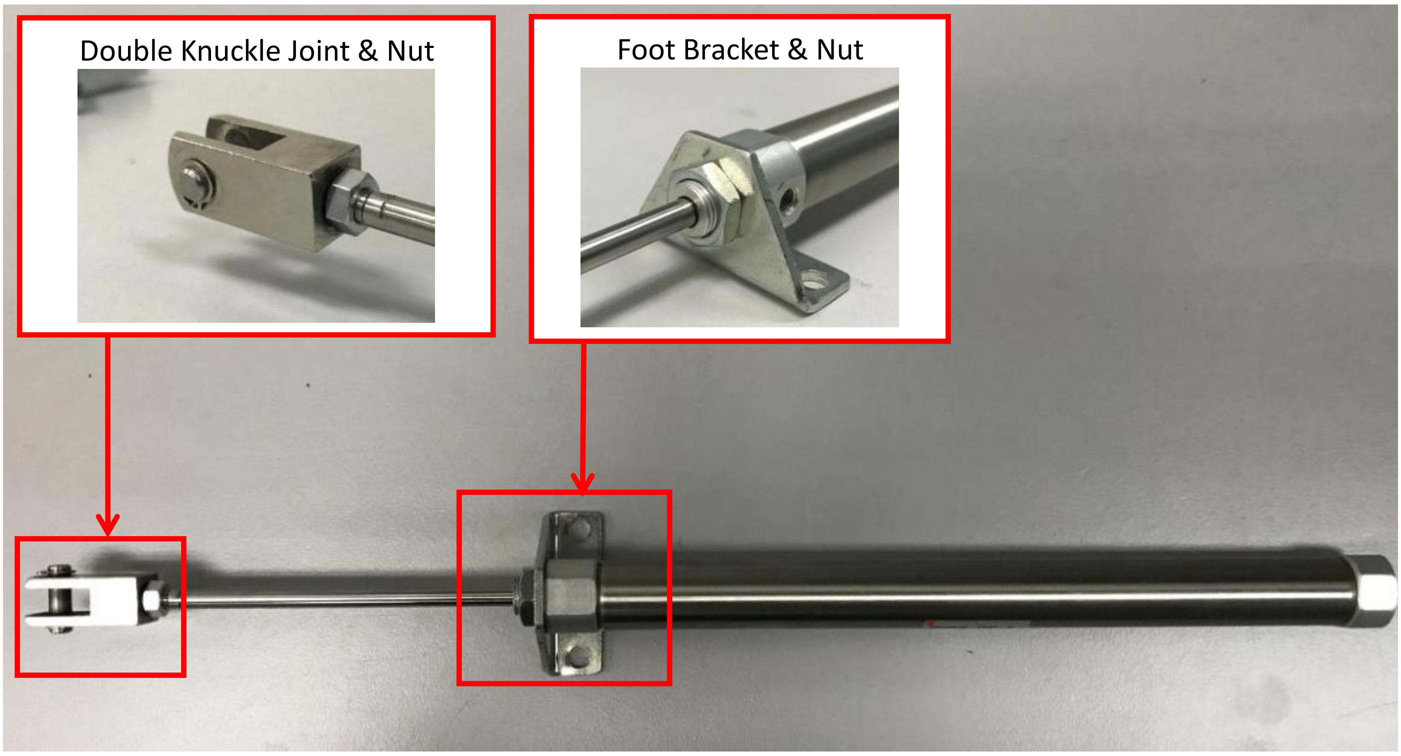

Figure 11 Inner Barrier Cylinder Module Assembly

3. Assemble Inner Barrier Cylinder Module as Figure 12. Tighten 2 x Screws.

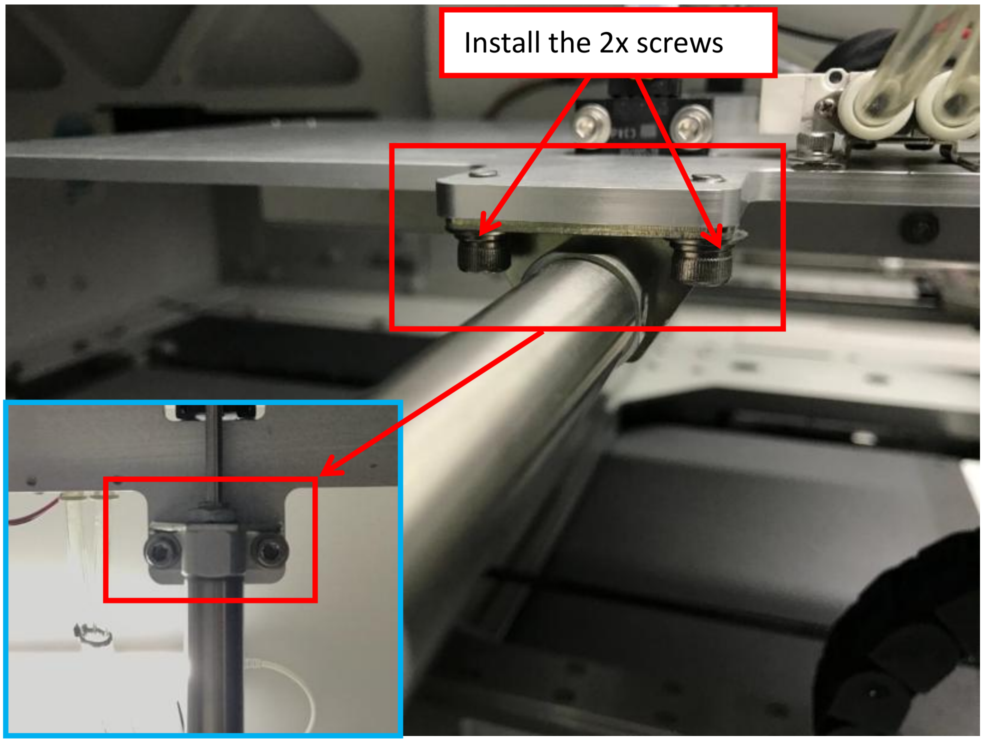

Figure 12 Inner Barrier Module

4. Gently slot Double Knuckle Joint into Drive Bar`s hook as shown in Figure 13

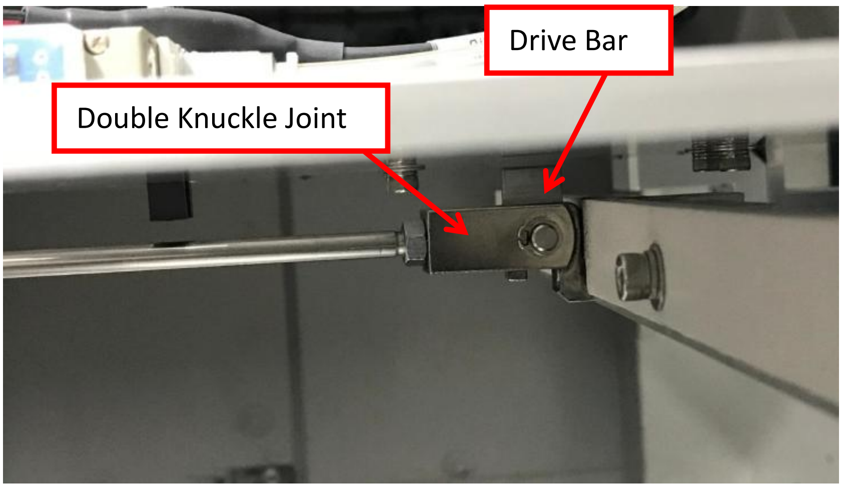

Figure 13 Double Knuckle Joint & Drive Bar

5. Assemble 2 x Speed Controller. Tighten both Speed Controller by using adjustable spanner as shown in Figure 14 & 15

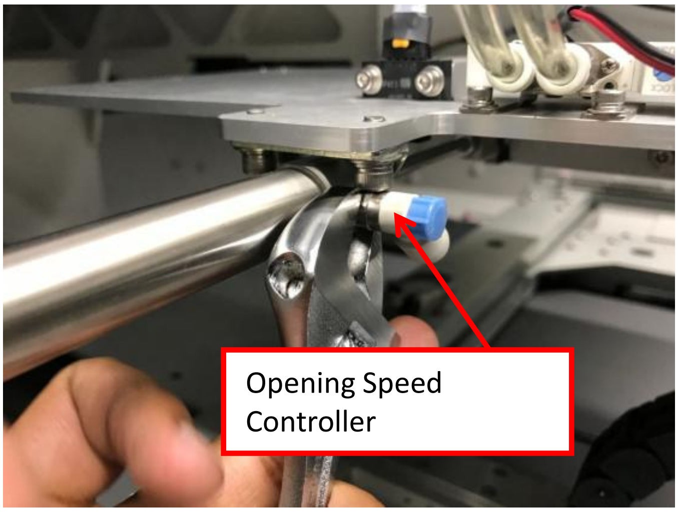

Figure 14 Opening Speed Controller Assembly

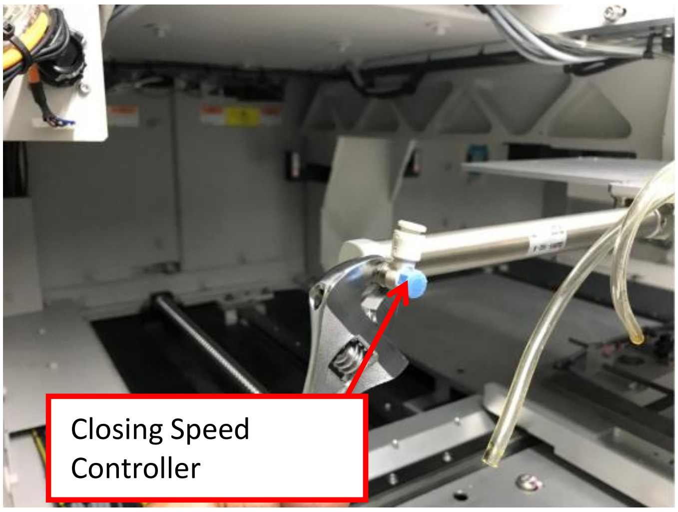

Figure 15 Closing Speed Controller Assembly

6. Gently plug 2 x Tubing respectively into Speed Controllers by refer to previous marking as below Figure 16

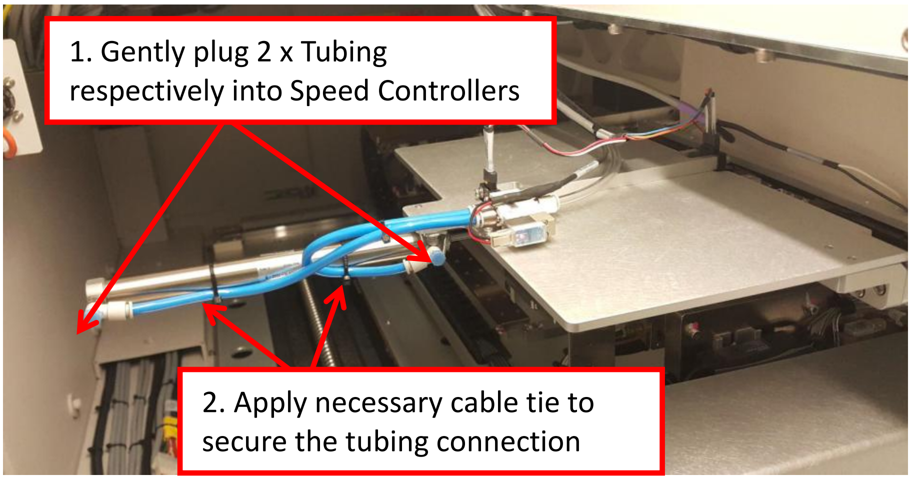

Figure 16 Tubing Connection

B. Inner Barrier Interlock Verification

1. Manually exercise Inner Barrier Cylinder where require to meet below condition :

   While Inner Barrier Cylinder fully extend,

   a) Inner Barrier shuttle plate MUST close firmly. (NO GAP)

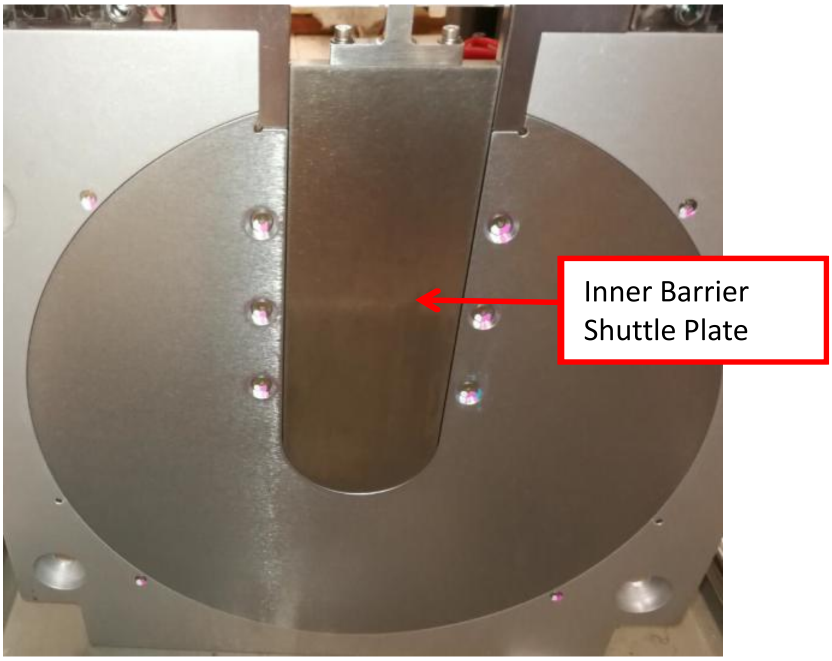

Figure 17 Inner Barrier Shuttle Plate Closing Condition

I. Perform below adjustment if gapping persist :

i. Loosen Nut at Double Knuckle Joint

ii. Re-adjust Inner Barrier Cylinder Front End Stroke

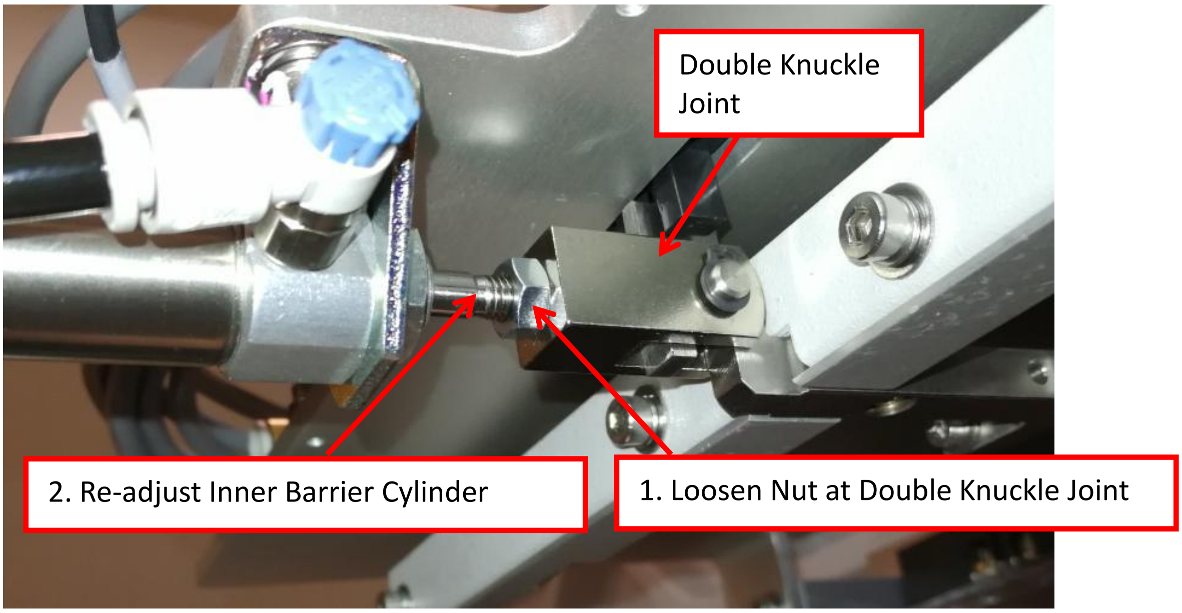

Figure 18 Inner Barrier Cylinder Front End Stroke Adjustment

iii. Tighten Nut at Double Knuckle Joint

iv. Visual Check Inner Barrier Shuttle Plate Closing Condition

b) Left Schmersal Switch & Right Schmersal Switch Connectivity MUST present where Inner barrier is closing firmly

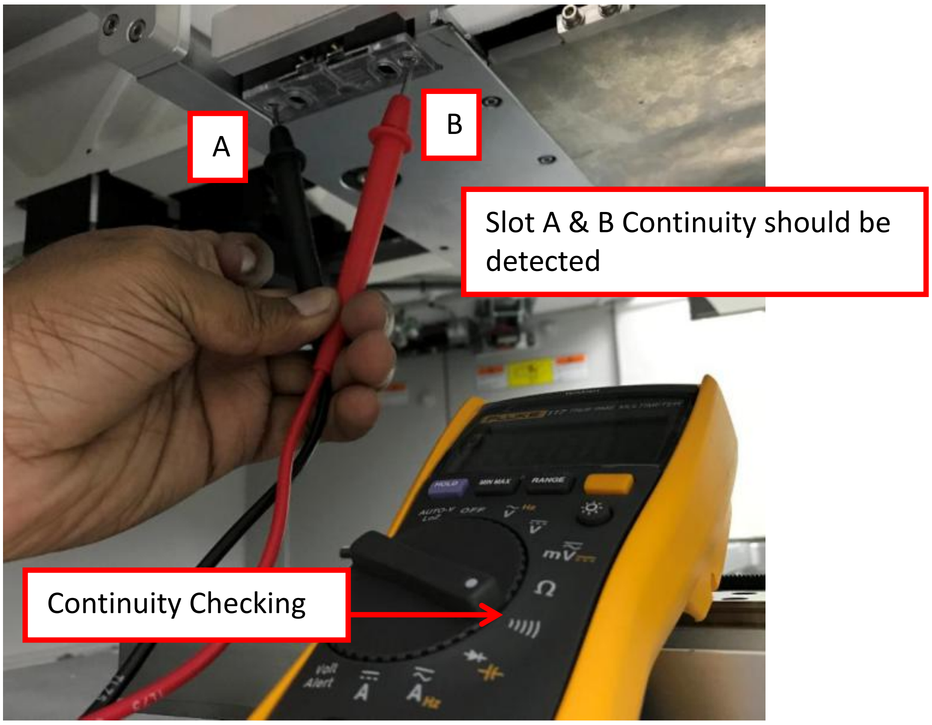

Figure 19 Continuity Checking (Left & Right Schmersal Switch)

I. Perform below adjustment if Left &Right Schmersal Switch Connectivity not present :

i. Loosen Nut at Double Knuckle Joint

ii. Re-adjust Inner Barrier Cylinder Front End Stroke

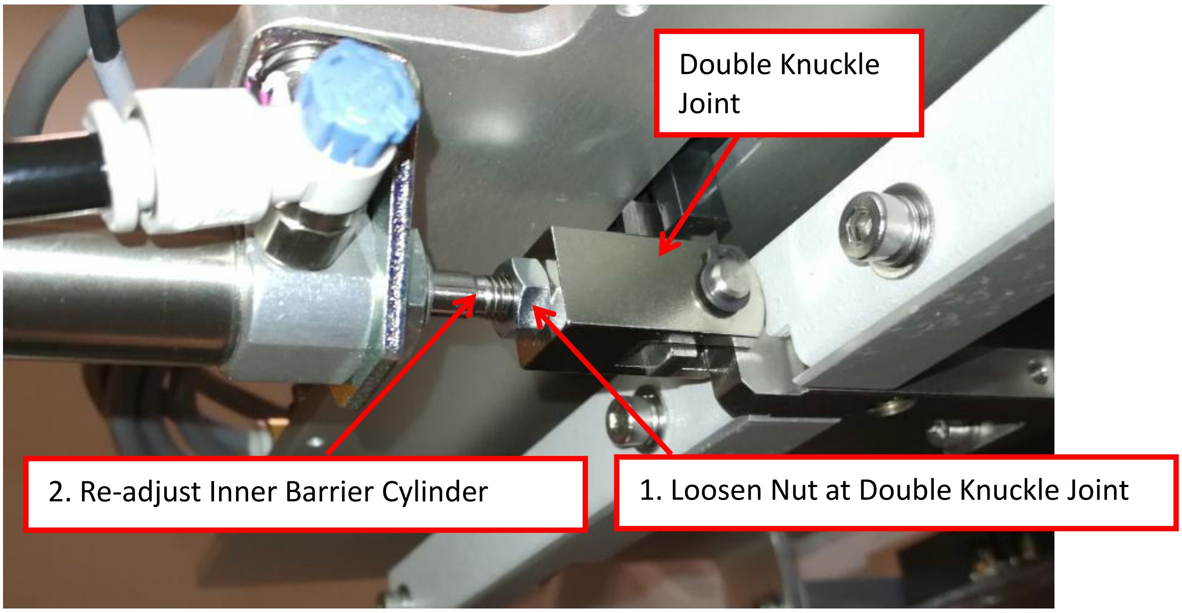

Figure 20 Inner Barrier Cylinder Front End Stroke Adjustment

iii. Tighten the Nut at Double Knuckle Joint

iv. Probe at Slot A and Slot B (Figure 21) again where Continuity should present upon inner barrier closing

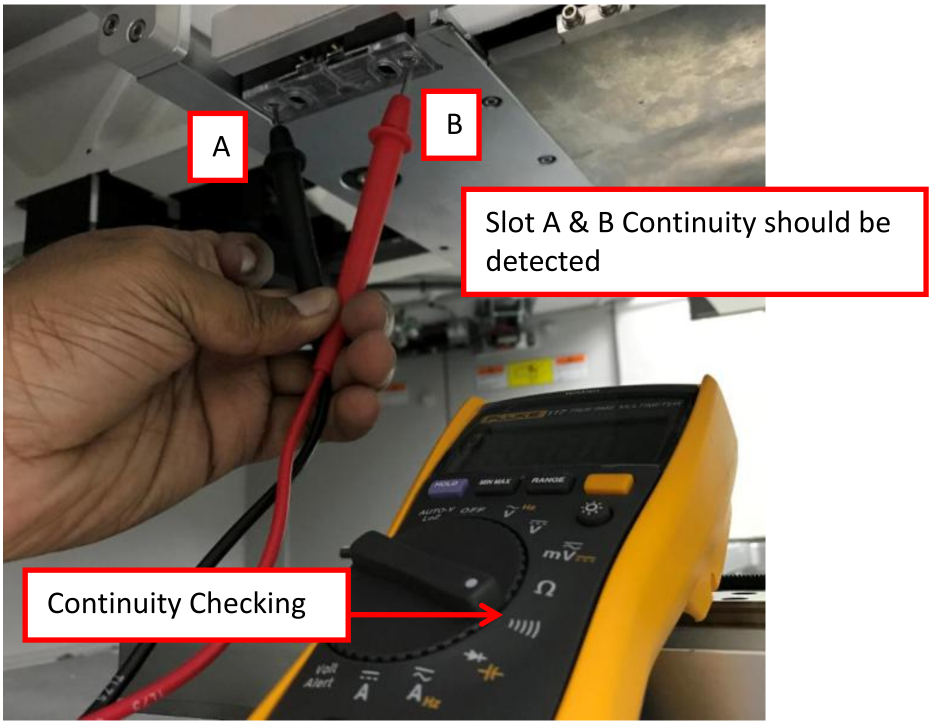

Figure 21 Continuity Checking (Left & Right Schmersal Switch)

C. Inner Barrier Opening & Closing Speed Adjustment

1. Turn on Air Pressure Regulator and set to 80 PSI
2. Close all access door
3. Open V810 GUI
4. At V810 GUI, go to Services -> Subsystem Status & Control -> Panel Handling and Click “Initialize” as shown in Figure 22

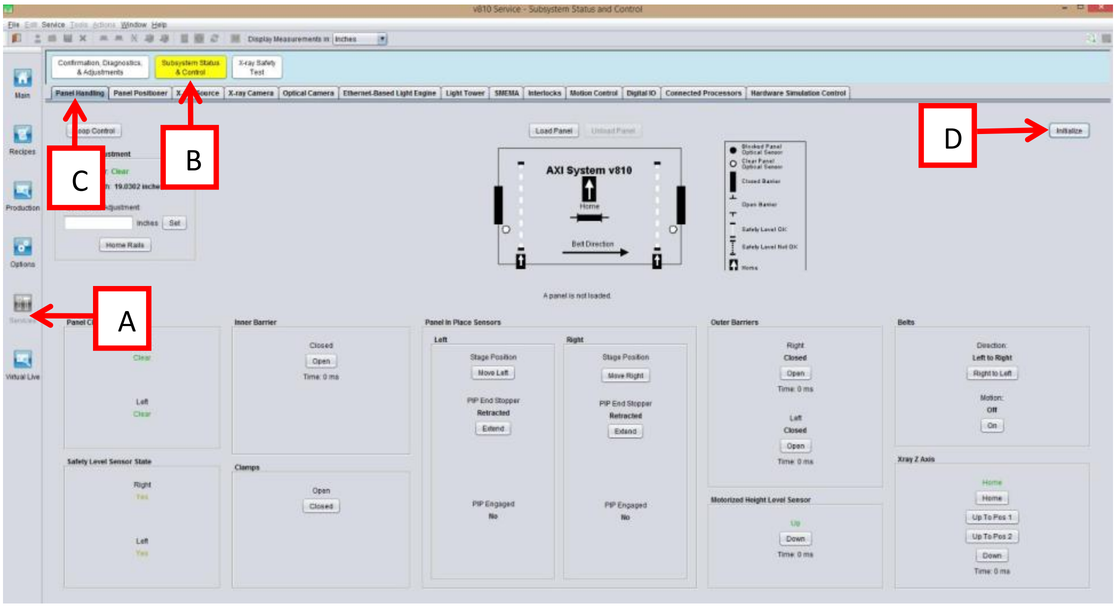

Figure 22 Panel Handling

5. Manually toggle inner barrier as below Figure 23

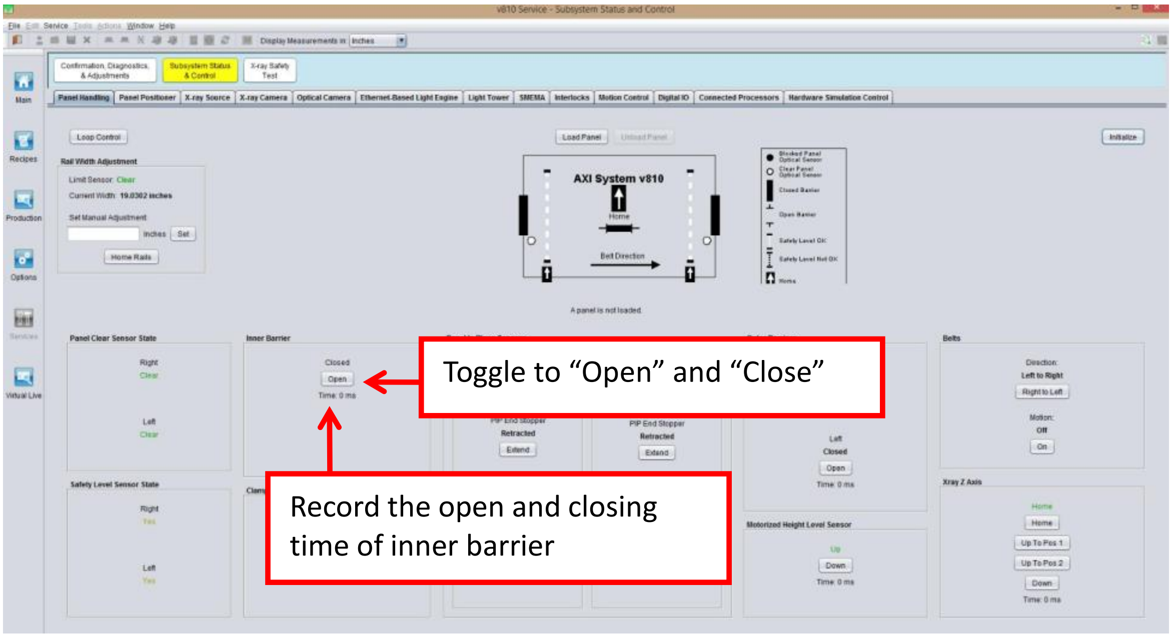

Figure 23 Panel Handling (Inner Barrier Speed)

6. Pull out speed control fitting knob, then adjust until Opening/Closing speed is within below range

Machine Model	Task	Speed (ms)
V810 STD with VM option	Inner barrier Opening & Closing	800 ~ 1000
V810 STD without VM option	Inner barrier Opening & Closing	600 ~ 750
V810 XXL	Inner barrier Opening & Closing	800 ~ 1000
V810 S2EX	Inner barrier Opening & Closing	1000 ~ 1200

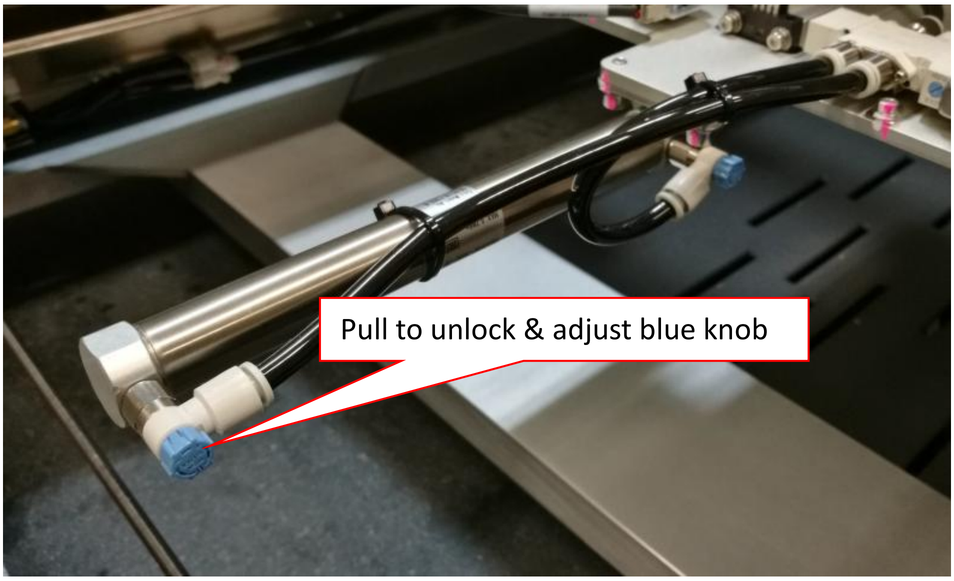

Figure 24 Inner barrier speed adjustment

7. Release Inner Barrier Speed Control Fitting Knob.Close Front doors.

8. Tun on X-ray sources & run 3 loops of CD&A.